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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2022-2023

Program/s: B Tech

Year: I Semester: II

Stream/s: Computer Science and Business Systems

Subject: Data Structures and Algorithms

Time: 3 hrs. (2:00pm to 5:00 pm)

Date: 27/June/ 2023

No. of Pages: 02

Marks: 100

Re - Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1_ is compulsory.
2) Out of remaining questions, attempt any _4_ questions.
3) **In all _5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO-2; BL-L	a.	Discuss an appropriate application of linked list.	[05]
CO-1; BL-L	b.	Classify data structures with suitable examples.	[05]
CO-1; BL-M	c.	How does looping affects the time complexity of an algorithm? Give suitable example.	[05]
CO-4; BL-M	d.	Explain significance of hashing with suitable example.	[05]
Q2 CO-2; BL-H	a	Convert and evaluate infixed expression: (a + b – c * (d + (e * 2))) using stack. Write all the necessary steps used during the conversion from infixed to prefixed expression.	[10]
CO-2; BL-M	b	How do you implement linear queue data structure using linked list? Perform one insert and one delete operation on the data structure with suitable algorithm / program.	[10]
Q3 CO-4; BL-M	a	Apply and explain selection sort on the following data set with worst case time complexity analysis. 15, 19, 12, 8, 23, 13, 45, 36, 78.	[10]

CO-2; BL-L	b	What do you mean by static data structure? What are its limitations? Demonstrate with suitable example.	[10]
Q4 CO-2; BL-H	a	Write the algorithm / function to insert and delete the element in circular queue with all the necessary conditions. Demonstrate this operation with suitable example.	[10]
CO-3; BL-M	b	What is the significance of binary search tree (BST)? Explain the possible delete operations performed on BST with suitable algorithms / functions.	[10]
Q5 CO-4; BL-L	a	Explain insertion sort algorithm with suitable example. Discuss its best case and worst case time complexity.	[10]
CO-3; BL-H	b	Consider the following message and encode it using Huffman encoding method: AAABBCBBACCCBBDDDDEEAABBB Also explain the use of binary tree in encoding the message.	[10]
Q6 CO-3; BL-H	a	Explain Dijkstra algorithm with suitable example (consider a graph with at least 8 nodes and 14 edges with appropriate values) to find the shortest path. Write all the necessary steps.	[10]
CO-3; BL-L	b	Explain inorder, preorder and postorder tree traversal algorithms using suitable example with at least 10 nodes and 15 edges in the tree.	[10]
Q7 CO-3; BL-M	a	Draw the graph with at least 8 nodes and 12 edges as an example. Apply depth first search method to traverse this graph and display the final sequence of traversal.	[10]
CO-2; BL-M	b	How do you implement a stack with linked list? Explain with suitable example and algorithm / program.	[10]